Lecture 22

Adjugates and equivalent descriptions

Uses.

Summary 13: Force columns add componentwise. A real vector's length normalises its direction; a scalar triple product gives signed volume.
Summary 18: Independence means that a zero linear combination has only zero coefficients. A basis is independent and spans its space.
Summary 19: Transposition reverses products and transposition twice restores a matrix. The matrix's columns specify its linear action.
Summary 20: Invertible square matrices are products of elementary matrices. Invertibility is equivalent to a zero null space and to unique solutions for every right side. The empty determinant is 1.
Summary 21: Row linearity and the row-operation rules are proved from the determinant definition. A determinant is nonzero exactly for invertible matrices, and determinants multiply.

Objectives. *The determinant is unchanged by transposition, so it can be expanded along any row or column. These expansions give the adjugate and Cramer's rule. Together with elimination, they describe the same invertible map through its equations, columns and determinant. A force-balance model supplies one application of the formulas.*

By the end of this lecture you will be able to:

- Prove transpose invariance and expansion along any row or column.
- Construct the adjugate and prove both adjugate product identities.
- Derive Cramer's rule and apply it only when the coefficient determinant is nonzero.
- Relate the equivalent square-matrix criteria and solve the stated two-cable equilibrium model.

22.1 Transposition and column rules

Let $K = \mathbb{R}$ or $K = \mathbb{C}$, with all determinants taken for square matrices of size $n \geq 1$ unless the empty convention is explicitly used.

Theorem 22.1 (Transpose invariance)**.** *For $A \in K^{n\times n}$, the determinant satisfies*

$$\det(A^{\mathsf{T}}) = \det A.$$

Proof. First consider an elementary matrix E. A row-exchange elementary matrix equals its transpose, as does a row-scaling matrix. Their transposes therefore have the same

determinants. A row-addition matrix has the entries of I_n with an extra scalar c in position (i, j), where $i \neq j$. Its transpose has that extra entry in position (j, i), so it is also a row-addition elementary matrix. Both determinants are 1 by the row-addition rule applied to I_n. Thus $\det(E^{\mathsf{T}}) = \det E$ for every elementary type.

If A is invertible, Lecture 20 writes it as a finite product $A = E_1 \cdots E_s$ of elementary matrices. Transposition reverses that order by Lecture 19, and multiplicativity gives

$$\det(A^{\mathsf{T}}) = \det(E_s^{\mathsf{T}}) \cdots \det(E_1^{\mathsf{T}}) = \det E_s \cdots \det E_1 = \det A.$$

The last equality commutes scalar factors, not matrix factors. For the empty product $A = I_n$, both determinants are 1.

If A is singular, its transpose is singular as well. Indeed, if A^{T} had inverse C, transposing both identities $A^{\mathsf{T}}C = CA^{\mathsf{T}} = I_n$ would give $C^{\mathsf{T}}A = AC^{\mathsf{T}} = I_n$ and make A invertible, a contradiction. Both determinants are therefore zero by Lecture 21's criterion. This proves the equality also in the singular case. ∎

Rows of A^{T} are columns of A. The theorem transfers every proved row rule to the corresponding column rule: the determinant is linear in each column separately, column swaps negate it, and adding a multiple of a different column preserves it. A zero column or equal columns gives determinant zero. For example, to scale one column, transpose, scale that row, apply the row rule and transpose back. Transpose invariance identifies the determinant before and after those two transpositions. The same argument gives each of the other column assertions with exactly its stated factor.

22.2 Expansion along any row or column

For $A \in K^{n\times n}$, let A_{ij} denote the submatrix obtained by deleting row i and column j. The corresponding *cofactor* is the scalar

$$C_{ij} = (-1)^{i+j} \det A_{ij}.$$

At size 1 the remaining matrix is empty, whose determinant is 1. Thus the sole cofactor of a 1×1 matrix is 1, even if its entry is zero.

Theorem 22.2 (Expansion formulas). *For any fixed column $j \in \{1, \ldots, n\}$ and fixed row $i \in \{1, \ldots, n\}$,*

$$\det A = \sum_{k=1}^{n} a_{kj} C_{kj}, \qquad \det A = \sum_{k=1}^{n} a_{ik} C_{ik}. \tag{22.1}$$

Each expansion uses one complete column or one complete row.

Proof. Move column j to the first position by $j-1$ adjacent column swaps, keeping the other columns in their relative order. Call the matrix B. Its determinant is $(-1)^{j-1} \det A$ by the column-swap rule. Expanding B along its first column gives

$$\det B = \sum_{k=1}^{n} (-1)^{k+1} a_{kj} \det A_{kj}.$$

The minors are exactly A_{kj}: deleting the moved column leaves every other column in its original relative order. Multiply the equation by $(-1)^{j-1}$. The sign in term k becomes $(-1)^{k+j}$, proving the column formula. This includes $j = 1$ with no swaps.

Apply the column formula to A^{T}. Its column i is row i of A, and its deleted submatrices are transposes of the corresponding deleted submatrices of A. Transpose invariance makes their determinants equal. The column formula for A^{T} is therefore the stated row formula for A. At size 1 both formulas use the empty determinant 1 and give the single entry, so that boundary case is included. ∎

To compute an expansion, choose a row or column with few nonzero entries. Keep the cofactor sign separate from the sign of its matrix entry. The definition and the theorem give the same determinant; choosing a new expansion does not introduce a new convention.

22.3 The adjugate and its two products

The *adjugate* of A is the transpose of its cofactor matrix:

$$\operatorname{adj} A = (C_{ij})^{\mathsf{T}}, \qquad (\operatorname{adj} A)_{ji} = C_{ij}. \tag{22.2}$$

The adjugate is different from the adjoint $A^{\dagger}$ of Lecture 19. The adjoint conjugates and transposes entries; the adjugate uses cofactors. Both have been defined also for singular square matrices.

Theorem 22.3 (Adjugate identities)**.** *For every $A \in K^{n\times n}$,*

$$A \operatorname{adj} A = \operatorname{adj} A\, A = (\det A) I_n. \tag{22.3}$$

If $\det A \neq 0$, its inverse is

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A. \tag{22.4}$$

Proof. The entry in row i, column j of $A \operatorname{adj} A$ is $\sum_k a_{ik} C_{jk}$. If $i = j$, this equals $\det A$ by expansion along row j. If $i \neq j$, replace row j of A by row i. The new matrix has two equal rows and determinant zero. Its cofactors along row j are still C_{jk}, because deleting that row removes the only changed entries. Expanding along row j therefore gives $\sum_k a_{ik} C_{jk} = 0$. This proves the first product identity entry by entry.

The entry in row i, column j of $\operatorname{adj} A\, A$ is $\sum_k C_{ki} a_{kj}$. If $i = j$, expansion along column i gives $\det A$. If $i \neq j$, replace column i by column j. The new matrix has equal columns and determinant zero. Its cofactors along column i remain C_{ki}, since their deletion removes the changed column. Expanding along that column gives $\sum_k C_{ki} a_{kj} = 0$. This proves the second product identity.

If $\det A \neq 0$, divide both product identities by this scalar. The proposed matrix in Eq. (22.4) satisfies both inverse identities, so it is the unique inverse by Lecture 19. ∎

For a complete computation, take

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

Its cofactor calculations are

$$\begin{aligned} C_{11} &= 1 - 2 = -1, & C_{12} &= -(1 - 0) = -1, & C_{13} &= 1 - 0 = 1, \\ C_{21} &= -(1 - 0) = -1, & C_{22} &= 2 - 0 = 2, & C_{23} &= -(2 - 0) = -2, \\ C_{31} &= 2 - 0 = 2, & C_{32} &= -(4 - 0) = -4, & C_{33} &= 2 - 1 = 1. \end{aligned}$$

Thus the cofactor matrix and its transpose are

$$(C_{ij}) = \begin{pmatrix} -1 & -1 & 1 \\ -1 & 2 & -2 \\ 2 & -4 & 1 \end{pmatrix}, \qquad \operatorname{adj} A = \begin{pmatrix} -1 & -1 & 2 \\ -1 & 2 & -4 \\ 1 & -2 & 1 \end{pmatrix}.$$

Expanding along the first row gives $\det A = 2(-1) + 1(-1) + 0 = -3$. It is nonzero, so the inverse is

$$A^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 & -2 \\ 1 & -2 & 4 \\ -1 & 2 & -1 \end{pmatrix}.$$

The adjugate identities give both inverse products, with the cofactor transpose fixing the position of every numerator. At size 1 the same construction says $\operatorname{adj}(a) = (1)$; the inverse formula is $(1/a)$ only when $a \neq 0$. When a determinant is zero, the adjugate products still hold but division by that determinant is unavailable.

22.4 Cramer's rule and its hypothesis

For $b \in K^n$, write $A[j \leftarrow b]$ for the matrix obtained by replacing column j of A by b, leaving every other column unchanged.

Theorem 22.4 (Cramer's rule). *If $A \in K^{n \times n}$ and $\det A \neq 0$, the unique solution of $Ax = b$ is*

$$x_j = \frac{\det A[j \leftarrow b]}{\det A}, \qquad j = 1, \dots, n.$$

Proof. The nonzero determinant guarantees a unique solution by Lectures 20–21. Write the columns of A as $a_1, \dots, a_n$. For that solution, the column-action formula gives $b = \sum_k x_k a_k$. Expand the determinant of $A[j \leftarrow b]$ by linearity in column j:

$$\det A[j \leftarrow b] = \sum_{k=1}^{n} x_k \det A[j \leftarrow a_k].$$

If $k \neq j$, that matrix has two equal columns and determinant zero. If $k = j$, it is A itself. Thus the sum is $x_j \det A$. Division by the assumed nonzero denominator gives the formula for every j. ∎

For singular systems, elimination is still required. Even zero values of every replacement determinant do not by themselves guarantee consistency. For example, take

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

The determinant of D and all three replacement determinants are zero: each replacement matrix retains at least one zero column. Nevertheless, the second equation of $Dx = b$ is $0 = 1$, a contradiction. Cramer's quotients do not apply at a zero denominator.

22.5 The equivalent descriptions

For square $A \in K^{n\times n}$, the results proved so far give the following equivalent descriptions:

matrix	A has a two-sided inverse
elimination	n pivots; reduced form I_n
equations	$\ker A = \{0_n\}$; $Ax = b$ uniquely solvable for every $b \in K^n$
columns	independent; spanning K^n; forming a basis of K^n
determinant	$\det A \neq 0$

The elimination, homogeneous-system, all-right-sides and determinant connections are Lectures 20–21. The column connections follow as well, without a general basis-cardinality theorem. A zero column combination is exactly $Ax = 0_n$, so column independence is equivalent to a zero null space. If A is invertible, $x = A^{-1}b$ expresses every target b as a column combination. Thus its columns span K^n.

Conversely, suppose the columns span K^n. Reduce A to $R = EA$ by an invertible product of elementary matrices. If R had a zero row in position q, choose $b = E^{-1}e_q$. Spanning would give a solution of $Ax = b$, and hence of $Rx = Eb = e_q$. Its row q would read $0 = 1$, a contradiction. Thus R has no zero row, has n pivots and is I_n. The inversion criterion proves that A is invertible. Finally, a basis means independence and spanning by Lecture 18, so the basis description is equivalent to the same conditions. The rank wording $\operatorname{rank} A = n$ uses the explicitly cited intrinsic-count convention from Lecture 18.

22.6 A force-balance model

A point at the origin is held by two straight cables leading to fixed anchors $P_1 = (-3, 4)$ and $P_2 = (5, 12)$ in metre coordinates. The cables exert nonnegative tensions T_1 and T_2, measured in newtons, towards their anchors. An applied force has column $F = (3, -15)$ in newtons. Use the equilibrium model that all force columns sum to zero.

Figure 22.1 fixes the directions before computing their components.

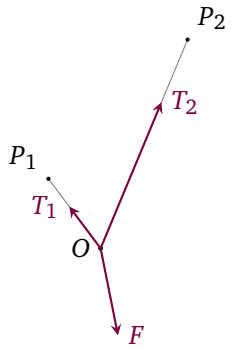


Figure 22.1: The cables pull towards their fixed anchors. The applied force points down and right. Force arrows show directions, with lengths separate from the position-coordinate scale and from the computed force magnitudes.

The unit direction columns, using lengths from Lecture 13, are

$$u_1 = (-3/5, 4/5), \qquad u_2 = (5/13, 12/13).$$

Force balance requires $T_1u_1 + T_2u_2 = -F$. Its component coefficient matrix and right side are

$$U = \begin{pmatrix} -3/5 & 5/13 \\ 4/5 & 12/13 \end{pmatrix}, \qquad b = \begin{pmatrix} -3 \\ 15 \end{pmatrix}.$$

The determinant is $-36/65 - 20/65 = -56/65 \neq 0$, so the balance equations have one solution. Cramer's numerators are

$$\det U[1 \leftarrow b] = \det \begin{pmatrix} -3 & 5/13 \\ 15 & 12/13 \end{pmatrix} = -\frac{111}{13}, \qquad \det U[2 \leftarrow b] = \det \begin{pmatrix} -3/5 & -3 \\ 4/5 & 15 \end{pmatrix} = -\frac{33}{5}.$$

The tensions are therefore

$$T_1 = \frac{555}{56}\ \text{N}, \qquad T_2 = \frac{429}{56}\ \text{N}.$$

Both values are positive, as required for cables pulling towards their anchors. The two cable forces have total horizontal component $-333/56 + 165/56 = -3$ newtons and total vertical component $444/56 + 396/56 = 15$ newtons. Adding $F = (3, -15)$ gives zero in each component. Thus the unique algebraic solution satisfies both the force balance and the physical nonnegative-tension restriction.

Summary 22.

- Theorem 22.1: transposition preserves the determinant. The row rules therefore hold for columns as well.
- Theorem 22.2: one complete row or column, with its cofactors $C_{ij} = (-1)^{i+j} \det A_{ij}$, gives the determinant.
- Theorem 22.3: the adjugate transposes the cofactor matrix and satisfies both product identities for every square A, including singular A. For $\det A \neq 0$, $A^{-1} = \operatorname{adj} A / \det A$.
- Theorem 22.4: nonzero determinant gives each solution coordinate as a replacement-column determinant divided by $\det A$. At a zero denominator, use elimination.
- § 22.5: invertibility, the square elimination and system criteria, a basis of columns, and nonzero determinant describe the same condition. In force models, algebraic solutions must also satisfy the stated physical restrictions.

Sheet 22 practices expansions, adjugates and Cramer's rule, including singular cases and the formal cross-product determinant. Lecture 23 returns to calculus and antiderivatives.

References

D. C. Lay, S. R. Lay and J. J. McDonald, *Linear Algebra and Its Applications*, 5th ed., Pearson, 2016, §§3.1–3.3. S. Lang, *Introduction to Linear Algebra*, 2nd ed., Springer, 1986, Chapter VII.